

YIFAN HOU

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Education

- **ETH Zürich** 2020 - 2026 (Expected in July)
Department of Computer Science
Ph.D. in Computer Science
Advisor: Prof. Mrinmaya Sachan & Prof. Antoine Bosselut
- **The Chinese University of Hong Kong (CUHK)** 2018 - 2020
Department of Computer Science and Engineering
M.Phil. in Computer Science and Engineering
Advisors: Prof. James Cheng & Prof. Ming-Chang Yang
- **Huazhong University of Science and Technology (HUST)** 2014 - 2018
School of Electronic Information and Communications
B.Eng. in Information Science for Advanced Class in Mathematics and Physics

Experience

- **Meta AI** — Research Intern October 2024 - February 2025
- **EPFL** — Visiting (Host: Prof. Antoine Bosselut) June 2023 - September 2023
- **EPFL** — Visiting (Host: Prof. Antoine Bosselut) June 2022 - September 2022
- **Tencent** — Research Intern June 2021 - September 2021
- **TTIC** — Remote Intern (Host: Prof. Mrinmaya Sachan) June 2020 - September 2020
- **NUS** — Visiting (Host: Prof. Richard T. B. Ma) June 2019 - September 2019
- **UIUC** — Summer Exchange July 2017 - August 2017

Research Philosophy

Interests: mechanistic interpretability, multimodal reasoning, transparent & reliable AI.

Brand: I pioneer **Transparent Multimodal AI**: bridging mechanistic interpretability, behavioral diagnostics, and algorithmic interventions to build foundation models that are provably reliable.

Philosophy: I aim to transition Multimodal Large Language Models (MLLMs) from brittle, black-box pattern matchers into *transparent and reliable world models*. My research systematically exposes multimodal illusions (e.g., knowledge shortcuts and extrapolation failures), uncovers their mechanistic roots within disjoint neural circuits, and develops algorithmic cures. Ultimately, my goal is to build uncertainty-aware, multimodal agents capable of mathematically verifiable reasoning for high-stakes applications and automated scientific discovery.

Selected Awards

- **Best Paper Award at MRL Workshop in EMNLP** 2022
- **SDSC Ph.D. Fellowship** 2022
- ICML Outstanding Reviewer 2022
- **Faculty Outstanding MPhil Thesis Award** (CUHK: Solo winner of Faculty of Engineering) 2020
- KDD Student Travel Award 2019
- Outstanding Graduates Awards (HUST) 2018
- National Scholarship 2016

Advising

Chengxi Li	Semester Project	2025-	Bachelor @ HKUST
Xingzhou Pang	Master Thesis	2025-	Master @ ETH Zürich
Yucheng Wang	Master Project	2025-2026	Master @ ETH Zürich
Yijie Tong	Master Thesis	2025-2026	Master @ UZH
Ziheng Chi	Master Thesis	2024-2025	Master @ ETH Zürich
Nayanika Debnath	Master Project	2024-2025	Master @ ETH Zürich
Monjoy Narayan Choudhury	Master Project	2024-2025	Master @ IIIT Bangalore
Yilei Tu	Master Project	2023-	ETH Zürich → PhD @ UBC
Buse Giledereli	Master Thesis	2023-2024	ETH Zürich → Data Scientist @ Prometeia
Yu Fei	Master Project	2022-2023	ETH Zürich → PhD @ UCI

Teaching Experience

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- TA to Prof. Fernando Perez-Cruz
Deep Learning (Autumn 2025)
Department of Computer Science, ETH Zürich
 - TA to Prof. Mrinmaya Sachan, Ryan Cotterell
Large Language Models (Spring 2023, Spring 2026)
Department of Computer Science, ETH Zürich
 - Head TA to Prof. Mrinmaya Sachan
Computational Semantics for Natural Language Processing (Spring 2021, 2022, 2023, 2024, 2025)
Department of Computer Science, ETH Zürich
 - TA to Prof. CHAN, Lai-Wan Lauren
Introduction to Social Networks (Spring 2020)
Department of Computer Science and Engineering, CUHK

Academic Service

Reviewer: ICML, NeurIPS, ICLR, ARR (ACL, EMNLP, NAACL), COLM

Selected Publications

Notations: My Mentee; * Equal contribution

Preprints

- [1] *Chimera: Diagnosing Shortcut Learning in Visual-Language Understanding.*
Ziheng Chi*, **Yifan Hou***, Chenxi Pang, Shaobo Cui, Mubashara Akhtar, Mrinmaya Sachan.
Manuscript, 2026.
- [2] *Stage-Wise Visual Component Decomposition for Diagram Understanding Enhancement.*
Junling Wang, **Yifan Hou**, Mrinmaya Sachan.
Manuscript, 2026.
- [3] *LLMs Find Their Way in Multi-Turn Conversation.*
Xingzhou Pang, Jiaoda Li, **Yifan Hou**, Ryan Cotterell, Mrinmaya Sachan.
Manuscript, 2026.

Conference Papers

- [1] *Unveiling the Visual Counting Bottleneck in Vision-Language Models.*
Xingzhou Pang*, **Yifan Hou***, Junling Wang, Mrinmaya Sachan.
ICML, 2026.
- [2] *Diversity Matters: Revisiting Test-Time Compute in Vision-Language Models.*
Yijie Tong*, **Yifan Hou***, Shaobo Cui, Antoine Bosselut, Mrinmaya Sachan.
ICML, 2026.
- [3] *Compose and Fuse: Revisiting the Foundational Bottlenecks in Multimodal Reasoning.*
Yucheng Wang*, **Yifan Hou***, Aydin Javadov, Mubashara Akhtar, Mrinmaya Sachan.
ICLR, 2026.
- [4] *Can Vision-Language Models Solve Visual Math Equations?*
Monjoy Narayan Choudhury*, Junling Wang*, **Yifan Hou**, Mrinmaya Sachan.
EMNLP (Short), 2025.
- [5] *HealthCards: Exploring Text-to-Image Generation as Visual Aids for Healthcare Knowledge Democratizing and Education.*
Qian Wu, Zheyao Gao, Longfei Gou, **Yifan Hou**, Ann Sin Nga Lau, Qi Dou
ACL, 2025.
- [6] *Do Vision-Language Models Really Understand Visual Language?*
Yifan Hou, Buse Giledereli, Yilei Tu, Mrinmaya Sachan.
ICML, 2025.
- [7] *What Do Language Models Learn in Context? The Structured Task Hypothesis.*
Jiaoda Li*, **Yifan Hou***, Mrinmaya Sachan, Ryan Cotterell.
ACL, 2024.
- [8] *Towards a Mechanistic Interpretation of Multi-Step Reasoning Capabilities of Language Models.*
Yifan Hou, Jiaoda Li, Yu Fei, Alessandro Stolfo, Wangchunshu Zhou, Guangtao Zeng, Antoine Bosselut, Mrinmaya Sachan.
EMNLP, 2023.
- [9] *Mitigating Label Biases for In-context Learning.*
Yu Fei, **Yifan Hou**, Zeming Chen, Antoine Bosselut.
ACL, 2023.
- [10] *Adapters for Enhanced Modeling of Multilingual Knowledge and Text.*
Yifan Hou, Wenxiang Jiao, Meizhen Liu, Carl Allen, Zhaopeng Tu, Mrinmaya Sachan.
EMNLP (Findings), 2022.
Best Paper at Multi-lingual Representation Learning Workshop.
- [11] *What Has Been Enhanced in my Knowledge-Enhanced Language Model?*
Yifan Hou, Guoji Fu, Mrinmaya Sachan.
EMNLP (Findings), 2022.
- [12] *Bird’s Eye: Probing for Linguistic Graph Structures with a Simple Information-Theoretic Approach.*
Yifan Hou, Mrinmaya Sachan.
ACL, 2021.
- [13] *Measuring and Improving the Use of Graph Information in Graph Neural Networks.*
Yifan Hou, Jian Zhang, James Cheng, Kaili Ma, Richard T. B. Ma, Hongzhi Chen, Ming-Chang Yang
ICLR, 2020.
- [14] *A Representation Learning Framework for Property Graphs.*
Yifan Hou, Hongzhi Chen, Changji Li, James Cheng, Ming-Chang Yang.
SIGKDD, 2019.